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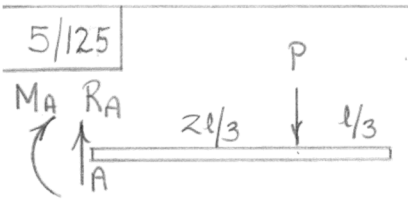
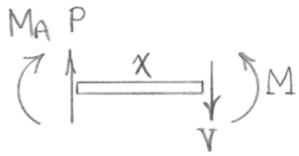


$+\uparrow \Sigma F = 0 \Rightarrow R_A = P$

$\circlearrowleft + \Sigma M_A = 0: \; -M_A - P\left(\frac{2l}{3}\right) = 0$

$M_A = -\frac{2Pl}{3}$

$0 < x < \frac{2l}{3}$:

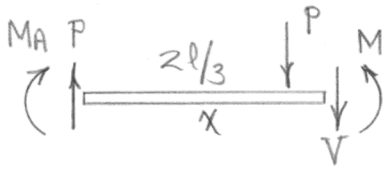


$+\uparrow \Sigma F = 0 \Rightarrow V = P$

$\circlearrowleft + \Sigma M = 0: \; \frac{2Pl}{3} - Px + M = 0$

$M = P\left(x - \frac{2l}{3}\right)$

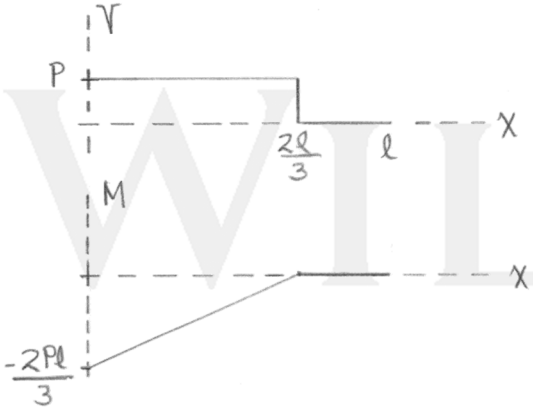
$\frac{2l}{3} < x < l$:



$+\uparrow \Sigma F = 0: \; P - P - V = 0, \; V = 0$

$\circlearrowleft + \Sigma M = 0: \; \frac{2Pl}{3} - P\left(\frac{2l}{3}\right) + M = 0$

$M = 0$



At $x = l/2$:

$V = P$

$M = P\left(\frac{l}{2} - \frac{2l}{3}\right)$

$= -\frac{Pl}{6}$